

Analysis of Alveolar Type II Cell Transcriptional States in Lethal COVID-19 Lung Biopsies

By Ashley Ramsawhook, PhD.

Introduction

This project is an independent analysis of the alveolar type II (AT2) cell transcriptional state in lethal COVID-19 biopsies based on the work by Melms *et al*, 2021, in their landmark paper “A Molecular Single-Cell Lung Atlas of Lethal COVID-19” (<https://www.nature.com/articles/s41586-021-03569-1>) While this notebook reproduces and modifies the pre-processing steps on the raw data performed by the authors, the exploration, analysis, deductions and conclusions are independent and self-directed. The agentic Large Language Model Claude Sonnet 5.0 (Anthropic) was utilised for code debugging, statistical methodology discussions and literature dataset navigation. This work was performed in Python 3.13.4. All code and markdown entries were written by myself.

The Dataset

The Melms *et al*, 2021 dataset consisted of lung biopsies extracted from live 7 healthy donors (not infected with COVID-19) and 19 deceased COVID-19 patients harvested post-mortem. Sequencing libraries were prepared from single nuclei suspensions. Only one biopsy was harvested per donor, making the terms “donor” and “sample” synonymous for this dataset. Cell Bender was implemented by the authors to remove contaminating ambient RNA and empty cells prior to the publication of the dataset.

Methodology

Manual Pre-processing

The raw counts dataset (GSE171524) was downloaded from Gene Expression Omnibus (GEO) using the link provided in the supplementary materials of the publication. For exploratory data analysis, raw counts for control and COVID-19 donor samples were loaded into Scanpy as an Anndata object and transposed to enable navigation of the data in a cell x genes format. Prior to doublet detection via SCVI tools “SOLO” model simulating R programming language’s “Seurat v3” functionality, nuclei were filtered to exclude genes with minimum coverage of 10 cells and subsequently subsetted to include only the top 2000 highly variable genes (HVGs). These steps minimised the probability of the deep-learning generative “SOLO” model from training on empty, burst or damaged cells and reduced the dataset dimensions and computational processing required for training. Following doublet identification, labelling and exclusion, mitochondrial and ribosomal genes were annotated. Genes with a coverage less than 3 nuclei were excluded. Scanpy quality control metrics (QC Metrics) were calculated to identify outliers in violin plots of `n_genes_by_count`, `total_counts`, percentage mitochondrial and ribosomal counts (`pct_counts_MT` and `pct_counts_ribo` respectively) depicted in fig 1. Nuclei with `n_genes_by_count` above the 98th percentile, with mitochondrial gene count proportions above 20% and

ribosomal gene count proportions above 5% were excluded as outliers as these nuclei were likely to contain stressed, burst or dying cells.

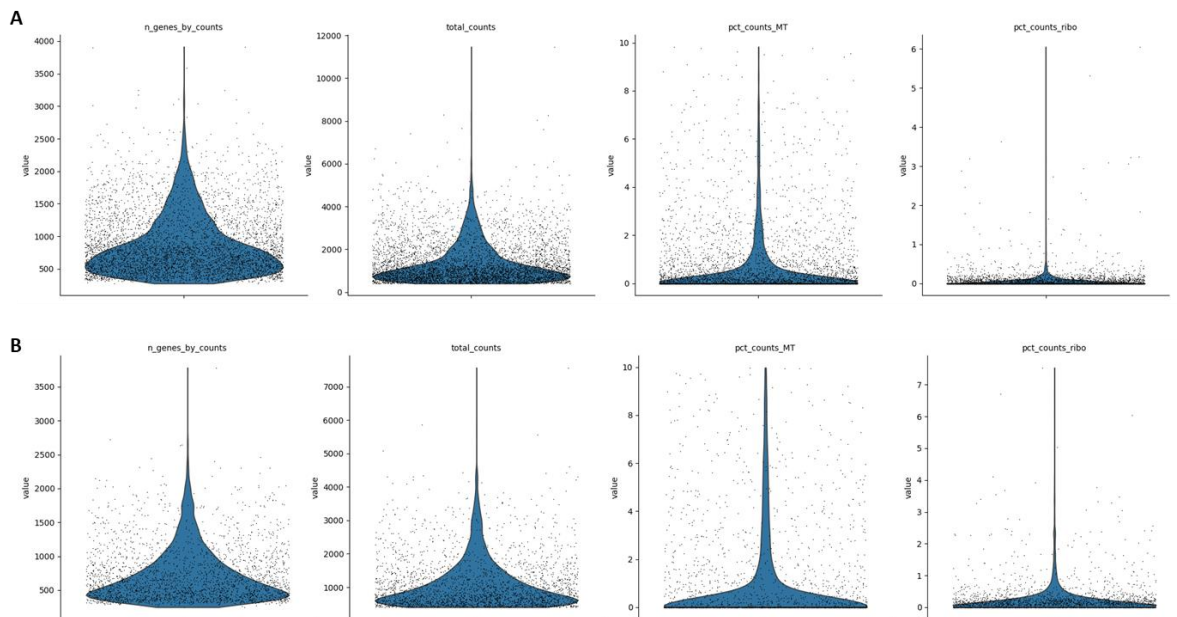


Figure 1. Violin plots of control (A) and COVID-19 (B) sample nuclei distributions by Scanpy QC metrics. From left to right the plots depict `n_genes_by_counts`, `total_counts`, `pct_counts_MT` and `pct_counts_ribo`. Nuclei above the 98th percentile of `n_genes_by_count`, > 20% `pct_counts_MT` and > 5% `pct_counts_ribo` were excluded as outliers.

Automated Pre-processing

The manual pre-processing steps implemented during the exploratory data analysis of single samples were compiled into a function and applied to all 26 samples in an automated workflow for concatenation into a single Anndata object as a sparse matrix to prevent tedious conversion post-normalisation. Raw read counts data was preserved in the “counts” layer of the Anndata object prior to normalisation to 1000 counts and log transformation was performed to reconstitute the count integrity. Gene filters were applied to reduce the dataset dimensions and exclude genes with less than 100 nuclei coverage and the top 3000 highly variable genes were saved without subsetting to include non-highly variable genes in downstream differential gene expression analysis. HGVs were saved in the Anndata “counts” layer. Sample integration to remove technical artefacts and batch effects between donors was conducted using SCVI tools with “sample” used as the batch key and the percentage ribosomal and mitochondrial counts as well as total counts being included as covariates.

Clustering

Latent representation, normalised expression and connectivity distances were extracted from the SCVI integration model and utilised for cell type clustering with the

uniform manifold approximation and projection (UMAP) algorithm. Cell cluster number label assignment to was performed using the Leiden algorithm. Cluster cell type marker assignment involved interrogation of significant ($\log_2$ fold change > 0.5 & p-value < 0.05) highly differentially expressed cluster genes, cross-referencing against Human Cell Atlas and PanglaoDB databases and validation against Cell Typist “Human Lung Atlas” cell type identification. Manually curated cluster label assignment was only updated against Cell Typist results if a complete mismatch in cell type and count existed. 25 Clusters were identified with multiple cell types distributed in more than one cluster (fig 2).



Figure 2. UMAP clusters of cell types by normalised expression. 25 clusters were identified and cell type labels were manually curated against Human Cell Atlas and PanglaoDB databases and corroborated via Cell Typist.

Single Cell Differential Expression

SCVI was employed to generate single cell differential expression using normalised counts from the previously trained model. Differential expression specific to alveolar type II (AT2) cells between healthy and COVID-19 conditions was extracted from the model, feature engineered to create a $\log_2$ fold change metric and filtered to select normalised mean counts with Bayes factor greater than 3 and $\log_2$ fold change greater than 0.5.

Pseudobulk Analysis

Pseudobulk profiles of differential gene expression were generated by building arrays of cell counts for each donor, integrating donor metadata, applying a minimum cell count threshold for donor and subsequently parsing this curated data into the PydeSeq2 pipeline for fitting onto the negative binomial distribution for statistical inference. Statistics were extracted from PydeSeq2, sorted by adjusted p-value (applying Benjamini-Hochberg Correction) and filtered for significance by screening for results with adjusted p-values less than 0.05 and $\log_2$ fold change greater than 0.5. Second layer Benjamini-Hochberg correction was performed when conducting

pseudobulk interrogation across multiple cell types simultaneously to prevent exponential compounding error incursion.

Gene Set Enrichment Analysis (GSEA)

GSEA was performed via GSEAPY and interrogated unfiltered pseudobulk results against *MSigDB Hallmark 2020* gene ontology vocabularies in pre-rank assessments, utilised significantly differentially expressed gene lists in enrichr over-representation analysis (ORA) against *KEGG 2021 Human* vocabularies and *Gene Ontology Biological Process 2021*. Significant gene lists were also selected for reactome analysis using the *Reactome 2022* vocabulary. Rankings were performed based on the Wald statistic generated by PydeSeq2, thus accounting for statistical power and effect magnitude.

Partition-based Graphical Abstraction (PAGA)

PAGA graph maps were constructed using Scanpy and build using Leiden algorithm partitions and K-nearest neighbour connectivities generated previously for UMAP clustering. Threshold was set to 0.1 to display moderately strong connections at minimum.

Results

Multiple Cell Types Populations are disrupted under COVID-19 Conditions

Cell counts across cell types were calculated and frequencies as a proportion of total cells per donor were compared between healthy and COVID-19 conditions (fig 3). The proportional representation of each cell type was compared, adjusting for multiple simultaneous testing errors at the single cell level using the Benjamini-Hochberg Correction.

Significant decreases in cell number were observed for AT1 ($p = 0.04$), AT2 cells ($p = 0.004$) and lung pericytes / alveolar fibroblasts ($p = 0.01$) in COVID-19 conditions compared to healthy controls. In contrast, monocytes / non-classical monocytes ($p = 0.0002$), alveolar macrophages ($p = 0.001$), peri-bronchiolar fibroblasts ($p = 0.002$), CD4 positive alpha-beta T cells ($p = 0.003$) and endothelial cells ($p = 0.006$) were significantly increased in COVID-19 conditions.

To further investigate the decrease in AT1 and AT2 cell types in COVID-19 conditions, differential gene expression was performed on these cell types at the single cell level via implementation of SCVI generative deep learning models.

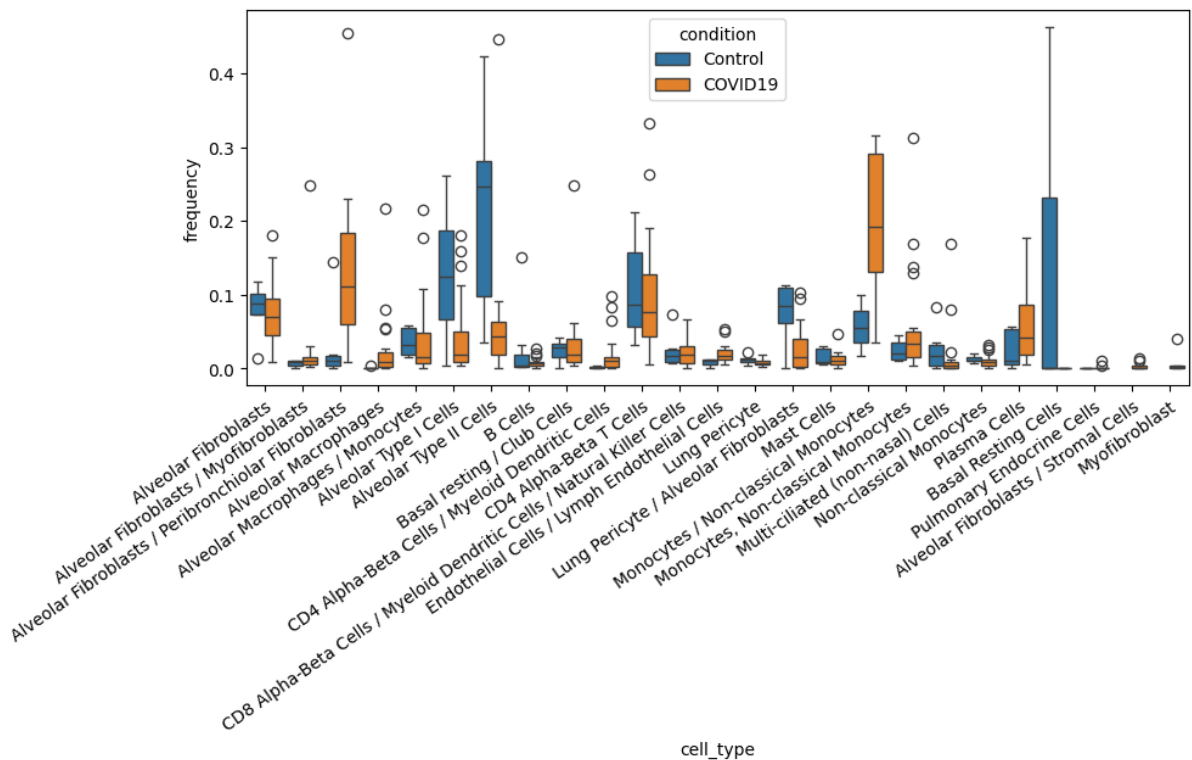


Figure 3. A Comparison of cell count frequencies per lung cell type between control healthy and COVID-19 conditions.

Circadian Rhythm, Cellular Exhaustion and Stress Response Genes Dominate the Significantly Differential Expression List in AT2 Cells at the Single Cell Level

The differentially expressed genes in AT2 cells depict a general trend of up-regulation under COVID-19 conditions relative to healthy controls (fig 4). 134 significantly differentially expressed genes were up-regulated in COVID-19 conditions compared to healthy controls. No genes were down-regulated in COVID-19. Multiple genes including *PHILDA3*, *EDA2R*, *CCNG1*, *ZMAT3* and *RRM2B* are components of the P53 signalling pathway and signal cellular stress. *LBR* and *MATR3* are involved in nuclear envelope architecture whereas *GABARAPL2* and *HSPA4L* relate to protein homeostasis as auto-phagosome and heat shock protein chaperones respectively. *RSP19* is a core structural component of ribosomes. The up-regulation of these proteins may indicate an immune response to trigger apoptosis in these cells under COVID-19 conditions. To determine whether these gene set observations were conserved at the donor-level, pseudobulk analysis was performed.

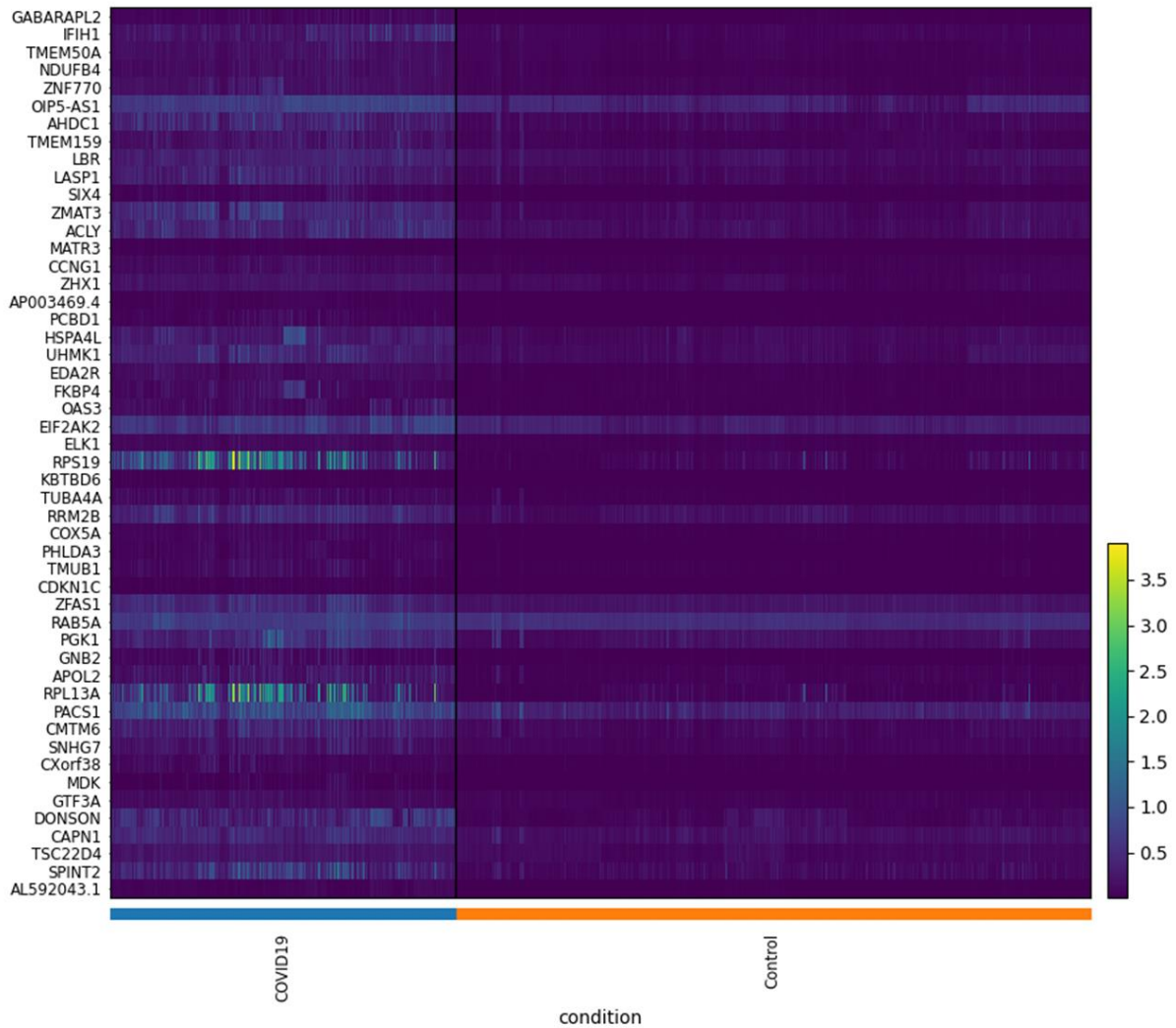


Figure 4. Heat map of the top 50 up-regulated significantly differentially expressed genes in COVID-19 conditions relative to healthy controls. *PHILDA3*, *EDA2R*, *CCNG1*, *ZMAT3* and *RRM2B* are components of the P53 signalling pathway.

Pseudobulk Analysis – Gene Set Enrichment Analysis (GSEA)

1258 differentially expressed genes were represented at the donor-level in the pseudobulk significant gene list consisting of 2392 entries (fig 5).

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
KCNK6	7.019782	7.132863	1.342226	5.314206	1.071234e-07	9.925839e-06
DONSON	10.135056	6.944506	1.017993	6.821763	8.992988e-12	6.754681e-09
GNAI2	5.217881	6.628590	1.391070	4.765103	1.887576e-06	8.861050e-05
S100A11	5.434327	6.323986	1.599679	3.953285	7.708559e-05	1.585142e-03
CXCL16	3.661063	6.258857	1.353419	4.624479	3.755415e-06	1.522543e-04
...	...	...	...	...	...	...
MYLK	4.529526	-4.145027	0.965535	-4.292986	1.762862e-05	5.021516e-04
IQCIN	15.234510	-4.176228	0.697130	-5.990603	2.090644e-09	4.661809e-07
FOSB	28.746254	-4.713729	1.119745	-4.209644	2.557735e-05	6.797288e-04
RIPOR3	11.540366	-4.867794	0.902500	-5.393678	6.902980e-08	7.203805e-06
PSTPIP1	5.432766	-5.891909	1.163115	-5.065630	4.070515e-07	2.806296e-05

2392 rows × 6 columns

Figure 5. Pseudobulk analysis of AT2 significant differential gene expression between COVID-19 and control conditions.

Unfiltered (non-significant) genes from the full pseudobulk differentially expressed results data frame were interrogated for representation in the *MSigDB Hallmark 2020* gene ontology vocabulary via the pre-rank module of the GSEAPY library (fig 6).

	Term	NES	NOM p-val	FDR q-val
0	mTORC1 Signaling	2.280246	0.000000	0.000000
1	Interferon Alpha Response	2.254218	0.000000	0.000000
2	Myc Targets V1	2.179700	0.000000	0.000000
5	Unfolded Protein Response	1.960320	0.000000	0.000170
4	p53 Pathway	2.018059	0.000000	0.000204
3	Oxidative Phosphorylation	2.124806	0.000000	0.000255
6	Interferon Gamma Response	1.944915	0.000000	0.000292
8	Pperoxisome	1.843219	0.000000	0.000568
7	Reactive Oxygen Species Pathway	1.858346	0.001333	0.000639
10	Cholesterol Homeostasis	1.801502	0.001267	0.000836

Figure 6. *MSigDB Hallmark 2020* rankings of enriched gene pathways within the full differentially expressed pseudobulk gene list.

The top hallmark gene sets from the gsea pre-rank assessment belonged to *mTORC1* signalling, Interferon Alpha Response, Myc Targets V1, Unfolded Protein Response

and the p53 Pathway as well as other pathways involved in oxidative stress responses. These pathways highlight a trend of cellular immune responses, survival signalling, programmed cell death and oxidative and metabolic stress. These pathways were all up-regulated in COVID-19 conditions relative to the controls.

To identify genes implicated in these pathways, the significantly differentially expressed genes were separated by direction of their effect size i.e. upregulated in COVID-19 conditions (positive) and upregulated in Control conditions (negative). Over Representation Analysis (ORA) was subsequently performed on up (fig 7) and down regulated (fig 8) gene sets.

Pseudobulk Analysis – Over Representation Analysis

ORA was performed using GSEAPY's enrichr module on significant differentially expressed genes from pseudobulk analysis against *KEGG 2021 Human* and *GO Biological Process 2021* vocabularies.

	Gene_set	Term	P-value	Adjusted P-value	Old P-value	Old adjusted P-value	Odds Ratio	Combined Score	Genes
0	KEGG_2021_Human	Ribosome	2.064318e-62	6.337456e-60	0	0	23.878989	3391.663618	RPL4;RPL5;RPL30;RPL3;RPL32;RPL31;RPL34;RPLP1;R...
1	KEGG_2021_Human	Coronavirus disease	1.479517e-49	2.271059e-47	0	0	11.441698	1286.446783	RPL4;RPL5;RPL30;RPL3;RPL32;RPL31;RPL34;RPLP1;R...
2	KEGG_2021_Human	Protein processing in endoplasmic reticulum	1.287587e-14	1.317630e-12	0	0	4.674325	149.500909	ERO1A;HSP90AB1;ATF6B;PRKCSH;HSPA4L;RPN1;HSP90B...
3	KEGG_2021_Human	Antigen processing and presentation	3.502201e-13	2.687939e-11	0	0	8.969649	257.251447	HSP90AB1;CTSS;B2M;HLA-DPA1;PDIA3;HSPA8;HSP90AA...
4	KEGG_2021_Human	Prion disease	1.288354e-10	7.910491e-09	0	0	3.178607	72.384785	NDUFA13;COX4I1;COX6A1;COX7C;PSMD8;TUBB8;CSNK2A...
...	...	...	...	...	...	...	...	...	...
4637	GO_Biological_Process_2021	regulation of generation of precursor metaboli...	9.797249e-01	9.806297e-01	0	0	0.251101	0.005143	NDC1
4638	GO_Biological_Process_2021	regulation of neurogenesis (GO:0050767)	9.842813e-01	9.849630e-01	0	0	0.235368	0.003729	B2M
4639	GO_Biological_Process_2021	monocarboxylic acid biosynthetic process (GO:0...	9.905532e-01	9.910105e-01	0	0	0.209146	0.001985	ACLY
4640	GO_Biological_Process_2021	double-strand break repair via nonhomologous e...	9.913219e-01	9.915507e-01	0	0	0.205332	0.001790	POLQ
4641	GO_Biological_Process_2021	inflammatory response (GO:0006954)	9.975440e-01	9.975440e-01	0	0	0.358516	0.000882	PRDX5;CEBPB;TUSC2;HSPG2;TNFRSF1A

4642 rows × 9 columns

Figure 7. ORA results of biological processes enriched in significantly up-regulated differentially expressed genes.

Interestingly, the ORA results illustrate Corona Virus disease response associated genes in the top 2 hits ($p = 2.27e-47$) and enrichment of genes associated with ribosomal construction pathways ($p = 6.33e-60$). These top two processes share redundancy in their enriched genes upregulated in COVID-19 conditions in AT2 cells. Additionally, the remaining three of the five hits relate to protein processing, antigen presentation and prion disease, commensurate with a cellular immune response.

	Gene_set	Term	P-value	Adjusted P-value	Old P-value	Old adjusted P-value	Odds Ratio	Combined Score	Genes
0	KEGG_2021_Human	Insulin resistance	0.000040	0.010446	0	0	4.003884	40.589328	PRKCE;NR1H2;PRKAG2;PRKAB1;NFKB1;MAPK10;RPS6KA3...
1	KEGG_2021_Human	Circadian rhythm	0.000133	0.017557	0	0	7.510692	67.033506	PER2;FBXW11;CUL1;CRY1;PRKAG2;RORA;PRKAB1
2	KEGG_2021_Human	Arrhythmogenic right ventricular cardiomyopathy	0.001264	0.097782	0	0	3.796343	25.334530	TCF7L1;CACNB4;LAMA1;TCF7;LMNA;CACNA2D3;CACNA2D...
3	KEGG_2021_Human	Adipocytokine signaling pathway	0.002130	0.097782	0	0	3.853185	23.702640	MAPK10;SOCS3;MAPK8;PRKAG2;PRKCQ;CD36;PRKAB1;NFKB1
4	KEGG_2021_Human	Oxytocin signaling pathway	0.002287	0.097782	0	0	2.675362	16.267161	MAP2K1;NPR1;CACNA2D3;ITPR1;CACNA2D2;PRKAG2;FOS...
...	...	...	...	...	...	...	...	...	...
2864	GO_Biological_Process_2021	mRNA processing (GO:0006397)	0.997987	0.998966	0	0	0.323125	0.000651	SFPQ;PCF11;RBM4B;SRSF7
2865	GO_Biological_Process_2021	negative regulation of cytokine production (GO...	0.997989	0.998966	0	0	0.158314	0.000319	RPS6KA5
2866	GO_Biological_Process_2021	gene expression (GO:0010467)	0.998199	0.998966	0	0	0.350899	0.000633	THRB;NUP98;SRSF7;GSPT1;ABCF1
2867	GO_Biological_Process_2021	rRNA metabolic process (GO:0016072)	0.998786	0.999169	0	0	0.146272	0.000178	MAPT
2868	GO_Biological_Process_2021	ncRNA processing (GO:0034470)	0.999720	0.999720	0	0	0.119657	0.000033	INTS12
2869 rows × 9 columns									

Figure 8. ORA results of biological processes enriched in significantly down-regulated differentially expressed genes.

The top 2 significantly downregulated processes were insulin resistance ($p = 0.010$) and circadian rhythm ($p = 0.017$). The enriched genes for the insulin resistance processes included members of the protein kinase C family, necrosis factor kappa Beta family, MAP Kinase signalling pathway family, *TBC1D4* of the *GLUT4* transporter regulation of insulin and *OGT* (O-GlcNAc) involved in insulin signalling. Lipid metabolism genes *NR1H2*, *CD36* and *SLC27A3* also featured in association with insulin resistance.

Pseudobulk – Reactome Analysis

Reactome pre-rank analysis was performed on the pseudobulk unfiltered differentially expressed genes against the *Reactome 2022* vocabulary. Results were split into “enriched” for up-regulated pathways (fig 9) and “depleted” for down-regulated ones (fig 10).

	Term	NES	NOM p-val	FDR q-val
0	Eukaryotic Translation Elongation R-HSA-156842	3.291909	0.0	0.0
1	Peptide Chain Elongation R-HSA-156902	3.283353	0.0	0.0
2	SRP-dependent Cotranslational Protein Targetin...	3.248776	0.0	0.0
3	Viral mRNA Translation R-HSA-192823	3.233536	0.0	0.0
4	Response Of EIF2AK4 (GCN2) To Amino Acid Defic...	3.227961	0.0	0.0
5	Selenocysteine Synthesis R-HSA-2408557	3.218916	0.0	0.0
6	Eukaryotic Translation Termination R-HSA-72764	3.210193	0.0	0.0
7	Nonsense Mediated Decay (NMD) Independent Of E...	3.208918	0.0	0.0
8	Selenoamino Acid Metabolism R-HSA-2408522	3.183585	0.0	0.0
9	Formation Of A Pool Of Free 40S Subunits R-HSA...	3.171082	0.0	0.0
10	Nonsense Mediated Decay (NMD) Enhanced By Exon...	3.141897	0.0	0.0
11	GTP Hydrolysis And Joining Of 60S Ribosomal Su...	3.127346	0.0	0.0
12	L13a-mediated Translational Silencing Of Cerul...	3.121908	0.0	0.0
13	Cap-dependent Translation Initiation R-HSA-72737	3.082041	0.0	0.0
14	Influenza Viral RNA Transcription And Replicat...	3.043410	0.0	0.0
15	Influenza Infection R-HSA-168255	3.016516	0.0	0.0
16	Regulation Of Expression Of SLITs And ROBOs R-...	3.005179	0.0	0.0
17	Cellular Response To Starvation R-HSA-9711097	3.002065	0.0	0.0
18	Major Pathway Of rRNA Processing In Nucleolus ...	2.938847	0.0	0.0
19	rRNA Processing In Nucleus And Cytosol R-HSA-8...	2.890560	0.0	0.0

Figure 9. Enriched biological pathways discovered via Reactome Analysis on the pseudobulk differential expression gene list.

The top 20 ranked enriched processes in COVID-19 conditions are dominated by those pertaining to protein synthesis, ribosome assembly, RNA processing, cellular immune response to infection and starvation.

	Term	NES	NOM p-val	FDR q-val
296	AURKA Activation By TPX2 R-HSA-8854518	1.548816	0.007323	0.049451
295	Metabolism Of Polyamines R-HSA-351202	1.549432	0.012966	0.049344
294	Nucleotide Salvage R-HSA-8956321	1.549963	0.026995	0.049230
293	Interleukin-12 Signaling R-HSA-9020591	1.550949	0.015023	0.048942
292	Regulation Of RAS By GAPs R-HSA-5658442	1.553683	0.012964	0.047905
291	Apoptotic Factor-Mediated Response R-HSA-111471	1.553733	0.025424	0.048058
290	Transcription Of E2F Targets Under Negative Co...	1.554058	0.025837	0.048090
289	Transferrin Endocytosis And Recycling R-HSA-91...	1.562371	0.020897	0.044482
288	Cross-presentation Of Soluble Exogenous Antige...	1.563958	0.013658	0.043955
286	FOXO-mediated Transcription R-HSA-9614085	1.568204	0.010003	0.042320
285	EPHA-mediated Growth Cone Collapse R-HSA-3928663	1.571896	0.022289	0.040932
284	RHOBTB2 GTPase Cycle R-HSA-9013418	1.573420	0.015684	0.040455
283	Immunoregulatory Interactions Between A Lympho...	1.573966	0.003706	0.040381
282	RHOJ GTPase Cycle R-HSA-9013409	1.578380	0.007137	0.038803
281	Regulation Of PLK1 Activity At G2/M Transition...	1.578524	0.003185	0.038885
280	Transport Of Bile Salts And Organic Acids, Met...	1.580687	0.009414	0.038195
279	Mitochondrial Biogenesis R-HSA-1592230	1.582158	0.004661	0.037762
278	Retrograde Transport At Trans-Golgi-Network R-...	1.583031	0.010359	0.037592
277	SARS-CoV-2 Activates/Modulates Innate And Adap...	1.583658	0.001928	0.037488
276	Post-chaperonin Tubulin Folding Pathway R-HSA-...	1.584099	0.019660	0.037468

Figure 10. Depleted biological pathways discovered via Reactome Analysis on the pseudobulk differential expression gene list.

The top ranked processes depleted under COVID-19 conditions pertain to cellular proliferation, DNA/RNA replication, cellular defence, immune signalling, apoptosis pathways, metabolic pathway signalling and protein synthesis and folding. Interestingly, cellular innate and adaptive immunity to SARs-CoV-2 infection is explicitly within the most depleted ranks.

To identify driver genes responsible for enrichment the up and down-regulated biological processes revealed via reactome analysis, lead genes were investigated within each pathway (fig 11).

	Term	NES	FDR q-val	Lead_genes
0	Eukaryotic Translation Elongation R-HSA-156842	3.291909	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
1	Peptide Chain Elongation R-HSA-156902	3.283353	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
2	SRP-dependent Cotranslational Protein Targetin...	3.248776	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
3	Viral mRNA Translation R-HSA-192823	3.233536	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
4	Response Of EIF2AK4 (GCN2) To Amino Acid Defic...	3.227961	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
5	Selenocysteine Synthesis R-HSA-2408557	3.218916	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
6	Eukaryotic Translation Termination R-HSA-72764	3.210193	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
7	Nonsense Mediated Decay (NMD) Independent Of E...	3.208918	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
8	Selenoamino Acid Metabolism R-HSA-2408522	3.183585	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
9	Formation Of A Pool Of Free 40S Subunits R-HSA...	3.171082	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
10	Nonsense Mediated Decay (NMD) Enhanced By Exon...	3.141897	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
11	GTP Hydrolysis And Joining Of 60S Ribosomal Su...	3.127346	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
12	L13a-mediated Translational Silencing Of Cerul...	3.121908	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
13	Cap-dependent Translation Initiation R-HSA-72737	3.082041	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
14	Influenza Viral RNA Transcription And Replicat...	3.043410	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
15	Influenza Infection R-HSA-168255	3.016516	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
16	Regulation Of Expression Of SLITs And ROBOs R-...	3.005179	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
17	Cellular Response To Starvation R-HSA-9711097	3.002065	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
18	Major Pathway Of rRNA Processing In Nucleolus ...	2.938847	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
19	rRNA Processing In Nucleus And Cytosol R-HSA-8...	2.890560	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
20	rRNA Processing R-HSA-72312	2.824635	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
21	Signaling By ROBO Receptors R-HSA-376176	2.820128	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
22	Translation R-HSA-72766	2.781837	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
23	SARS-CoV-2 Modulates Host Translation Machiner...	2.764398	0.0	RPS19;RPS11;RPS17;RPS24;RPS5;RPS27A;RPS3;RPS29...
24	Formation Of Ternary Complex, And Subsequently...	2.643369	0.0	RPS19;RPS11;RPS17;RPS24;RPS5;RPS27A;RPS3;RPS29...
25	Translation Initiation Complex Formation R-HSA...	2.603711	0.0	RPS19;RPS11;RPS17;RPS24;RPS5;RPS27A;RPS3;RPS29...
26	Ribosomal Scanning And Start Codon Recognition...	2.601839	0.0	RPS19;RPS11;RPS17;RPS24;RPS5;RPS27A;RPS3;RPS29...
27	mRNA Activation Upon Binding Of Cap-Binding Co...	2.573667	0.0	RPS19;RPS11;RPS17;RPS24;RPS5;RPS27A;RPS3;RPS29...
28	Metabolism Of Amino Acids And Derivatives R-HS...	2.539536	0.0	RPL13A;RPS19;RPL28;RPL36;RPS11;RPS17;RPL7A;RPS...
29	SARS-CoV-2-host Interactions R-HSA-9705683	2.348179	0.0	HLA-C;RPS19;B2M;RPS11;RPS17;RPS24;RPS5;RPS27A;...

Figure 11. Lead gene identification across enriched biological pathways revealed via reactome analysis.

While many of the top 30 pathways relate to ribosome assembly and functioning, protein synthesis via translation and host cell immune response, viral mRNA translation, influenza and viral transfection and SARs-Cov-2-host interaction pathways are also represented. Redundancy of ribosomal subunit gene enrichment exists for all of these pathways but *HLA-C* and *B2M* genes stand out for SARs-CoV-2-host interactions.

Interrogation of P53 Signalling Pathway Gene Distribution in AT2 Cells

ZMAT3 and *PHLDA3* are key components of canonical P53 signalling pathways. They are downstream targets of P53 and transcriptionally activated via direct binding of P53 to their promoters. *ZMAT3*, an RNA-binding protein positively regulates p53 expression through exon-exclusion and hence preclusion of P53 suppressor activation (Bieging-Rolett *et al*, 2021). *PHLDA3* activation results in suppression of *AKT* signalling in AT2 cells, alleviating inhibition of AT2 progenitor transdifferentiation to AT1 cells (Ohnishi *et al*, 2024). Considering the magnitude of downregulation of *ZMAT3* (Bayes Factor= 4.14, LFC mean = -2.55) and *PHLDA3* (Bayes Factor= 3.63 and LFC mean = -3.43) in differential expression at the single cell level and the conservation of P53 signalling pathway gene enrichment in pseudobulk and Over Represent Analysis, *ZMAT3* and *PHLDA3* distribution was interrogated at the single cell and donor levels (figs 12 and 13 respectively).

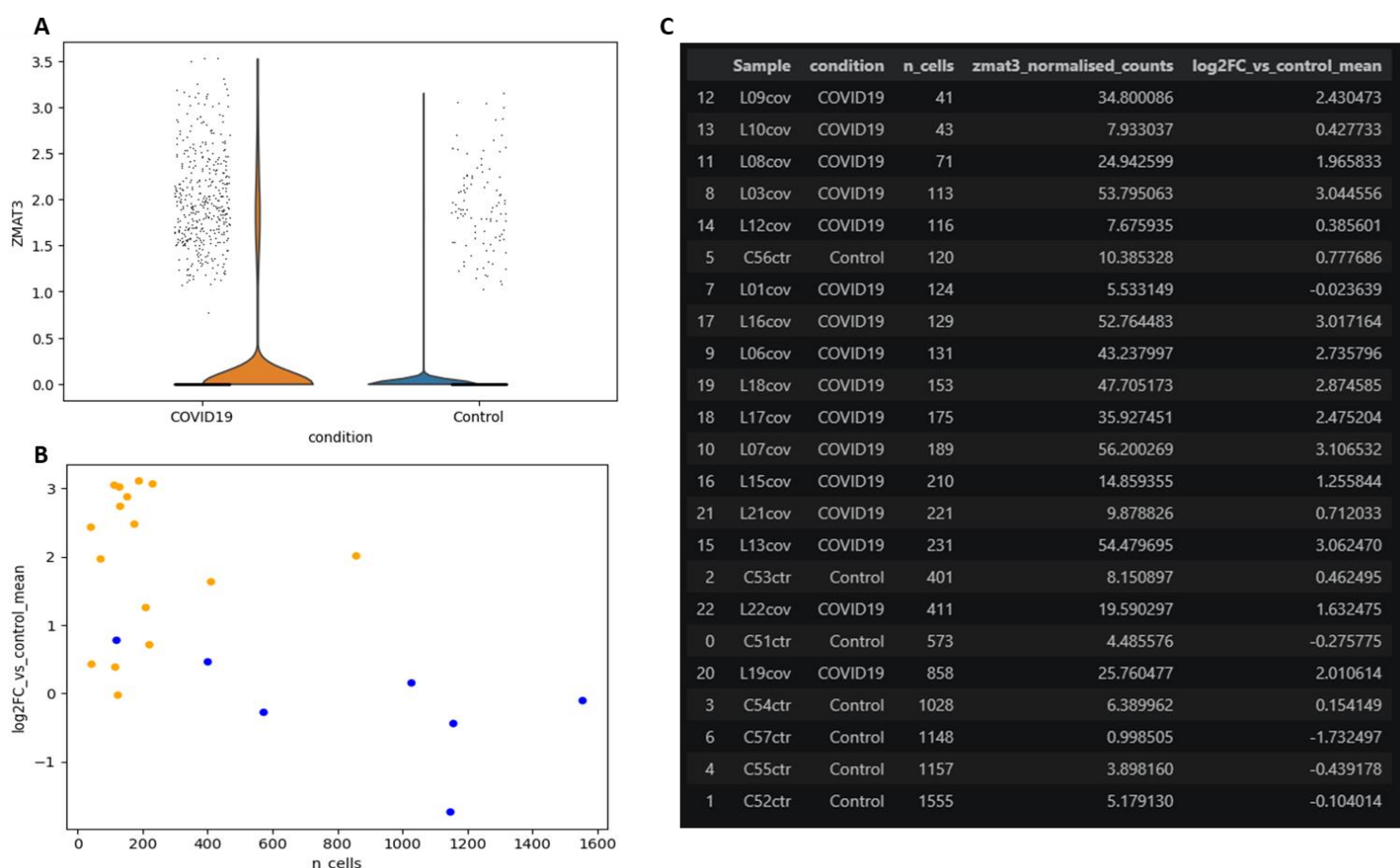


Figure 12. (A) Violin plot of *ZMAT3* expression distribution within single AT2 cells between conditions. (B) Scatterplot of AT2 cell *ZMAT3* log2 fold change with increasing cell donor cell numbers. (C) AT2 cell counts per donor vs *ZMAT3* log2 fold change in expression.

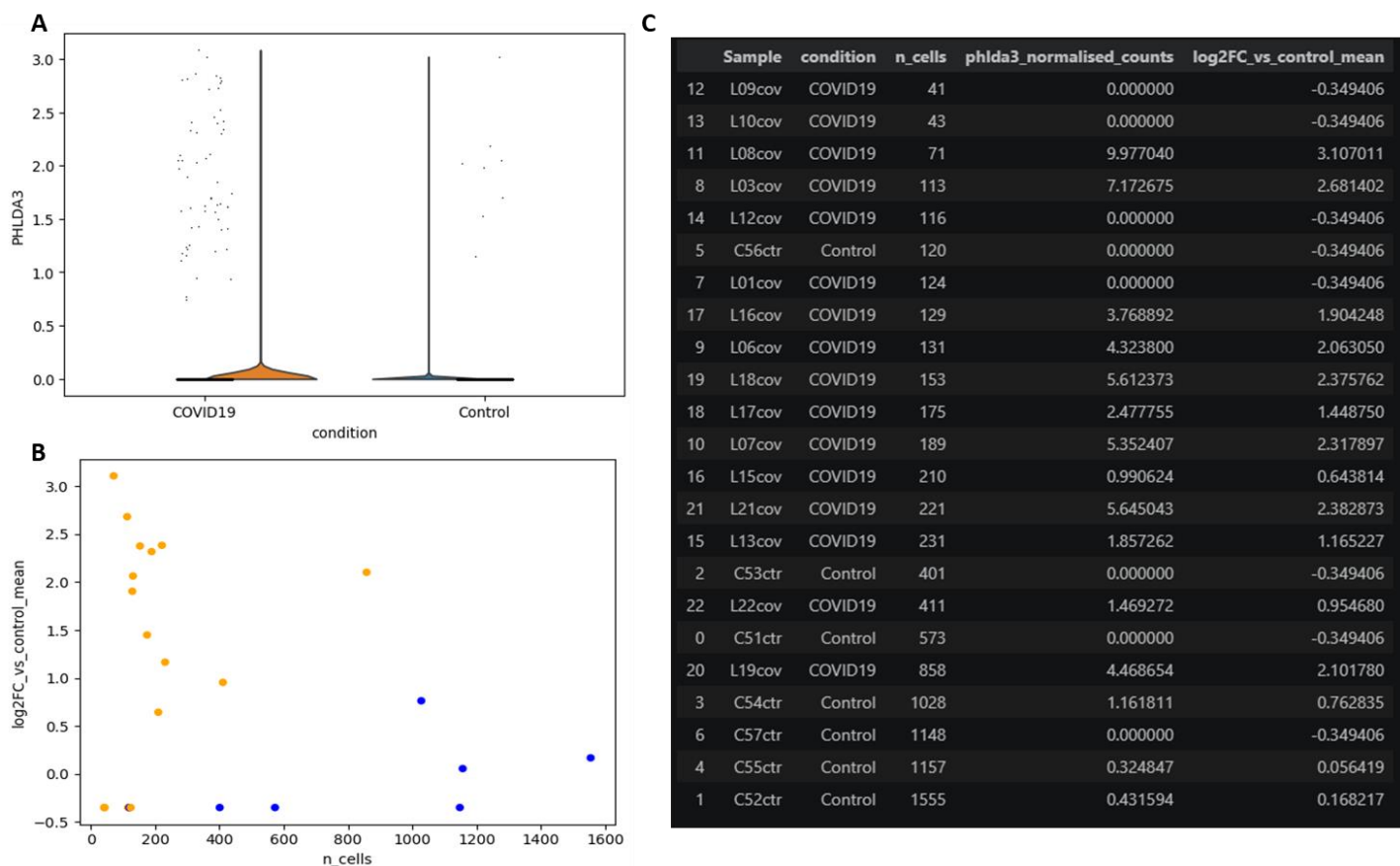


Figure 13. (A) Violin plot of *PHLDA3* expression distribution in single AT2 cells. (B) Scatterplot of AT2 *PHLDA3* log2 fold change in expression with increasing donor cell number. (C) AT2 cell counts per donor vs *PHLDA3* log2 fold change in expression.

Two subsets of *ZMAT3* expression exist in the AT2 cell population in both COVID-19 and control conditions (fig 12A). The majority of cells have no *ZMAT3* expression in either condition, though there is a greater preponderance of cells in COVID-19 conditions. In COVID-19 conditions, the smaller subset of AT2 cells demonstrate a diffuse pattern of expression with majority of cells expressing *ZMAT3* between 1 and 2.5 and fewer cells expressing above 2.5 to 3.5 fold levels. In contrast, the *ZMAT3* expression in control AT2 cells within smaller subpopulation is approximately uniform in cell density (illustrated by the straight vertical line with no bulge) and *ZMAT3* expression ranges between 1 and approximately 3-fold.

Analysis of the *ZMAT3* expression in AT2 cells between donors (fig 12B) revealed that *ZMAT3* expression is not independent of cell number (Pearson Correlation Coefficient, $r = 0.593$, $p = 0.003$), suggesting a moderate positive correlation in *ZMAT3* expression as donor cell numbers increase. However control donors appear, visually, to be decreasing in *ZMAT3* expression level as donor cell numbers increase.

Similar to *ZMAT3* expression in AT2 cells, a subset of *PHLDA3* expressing cells exist within a majority of *PHLDA3* non-expressing cells in both COVID-19 and control

conditions (fig 13A), with the former possessing greater numbers of expressing and non-expressing cells than the latter. In COVID-19 conditions, the *PHLDA3* expression ranges between 0.75 to over 3 while in control cells expression is narrower between 1 and 2.2 for the main expressing cluster and 1 possible outlier expressing above 3.

A weak, insignificant, negative correlation between donor cell number and *PHLDA3* expression exists. A clear disparity in *PHLDA3* expression is evident between control donors as a subset (n=3) do not express *PHLDA3* regardless of increasing cell number are distinct from *PHLDA3* expressing donors and have identical expression levels to a small subset (n=2) of COVID-19 donors. Interestingly, control donors on average exhibit lower *PHLDA3* expression despite possessing more cells per donor than COVID-19 donors.

For both *ZMAT3* and *PHLDA3* expression distribution in AT2 cells, the juxtaposition of single cell violin plots and donor level scatterplots against donor cell counts (fig 12C & 13C) revealed that subsets of expressing and non-expressing *ZMAT3* and *PHLDA3* cells exist *within* each donor population, not segregated binary populations of expression or non-expressing donors, as the violin plots alone seem to suggest.

Analysis of P53 Target Gene Expression Distribution across Lung Cell Types

A list of 343 census genes involved in the P53 signalling pathway were sourced from Fischer, 2017 and examined for their expression levels and distribution across all lung cell types and the results were filtered for statistical significance into up-regulated (fig 14A) and down-regulated (fig 14B) lists.

Up-regulated

A

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	cell_type	gene	padj_across_celltypes
283	73.105281	5.308973	0.705307	7.527180	5.184799e-14	9.874044e-11	Alveolar Type II Cells	RPS19	9.851117e-13
591	30.450933	7.370949	1.036825	7.109152	1.167579e-12	3.012076e-09	Alveolar Type I Cells	RPS19	1.109200e-11
309	20.187810	3.088713	0.431307	7.161279	7.992801e-13	3.012076e-09	Alveolar Type I Cells	ZMAT3	1.518632e-11
315	18.160499	2.351939	0.361511	6.505861	7.724957e-11	6.783824e-08	Alveolar Type I Cells	RRM2B	1.467742e-09
313	18.160499	2.351939	0.361511	6.505861	7.724957e-11	6.783824e-08	Alveolar Type I Cells	RRM2B	1.467742e-09
1	23.242237	2.583013	0.421443	6.128978	8.844538e-10	2.427315e-07	Alveolar Type II Cells	ZMAT3	8.402311e-09
213	6.305758	3.019077	0.525759	5.742321	9.338780e-09	1.687009e-06	Alveolar Type II Cells	ADGRG1	1.774368e-07
16	14.218205	1.857983	0.324125	5.732307	9.907399e-09	1.767356e-06	Alveolar Type II Cells	RPS27L	1.882406e-07
2106	19.195095	3.741472	0.659730	5.671217	1.417864e-08	5.054180e-06	Alveolar Macrophages / Monocytes	RGL1	2.693942e-07
3	4.136906	2.708720	0.482283	5.616455	1.949151e-08	2.956976e-06	Alveolar Type II Cells	EDA2R	3.703387e-07

B

Down-regulated										
	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	cell_type	gene	padj_across_celltypes	
	577	68.913149	-1.560165	0.256616	-6.079768	1.203569e-09	5.401534e-07	Alveolar Type I Cells	P3H2	2.166424e-08
	2533	3.558848	-3.651589	0.609330	-5.992796	2.062636e-09	9.577193e-07	Alveolar Fibroblasts	SYTL1	3.919008e-08
	270	169.915941	-2.319581	0.403135	-5.753859	8.722892e-09	1.616680e-06	Alveolar Type II Cells	P3H2	7.850603e-08
	2887	11.296333	-2.480592	0.496307	-4.998096	5.789911e-07	1.757317e-04	Endothelial Cells / Lymph Endothelial Cells	LMNA	1.100083e-05
	913	99.820388	-0.709901	0.142795	-4.971456	6.645197e-07	8.064853e-05	Monocytes / Non- classical Monocytes	TRAK1	1.262587e-05
	3201	97.757171	-1.279755	0.271338	-4.716456	2.399886e-06	1.807318e-04	Monocytes, Non- classical Monocytes	RNF144B	2.279892e-05
	2073	23.566586	-1.120193	0.236956	-4.727440	2.273680e-06	3.025813e-04	Alveolar Macrophages / Monocytes	GBE1	4.319992e-05
	2630	11.982758	-2.395091	0.520754	-4.599275	4.239643e-06	4.330796e-04	Alveolar Fibroblasts	ABTB2	8.055322e-05
	857	408.564239	-0.908691	0.199260	-4.560331	5.107299e-06	3.428167e-04	Monocytes / Non- classical Monocytes	FCHSD2	9.703868e-05
	2555	8.817040	-1.222920	0.272668	-4.485021	7.290682e-06	6.258346e-04	Alveolar Fibroblasts	INPP1	1.385230e-04

Figure 14. Cell type distribution and expression levels by log2 fold change of statistically significant census P53 target genes across the lung. Panel **A** depicts up-regulated genes and panel **B** depicts down-regulated genes relative to healthy controls.

263 significantly up-regulated hits were discovered within the P53 target genes across multiple lung cell types. Notably, of the top 10 recorded cell types, 5 pertained to AT2 cells and 4 to AT1 cells, with ribosomal *RSP19* and *ZMAT3* included in the top 3 hits. Alveolar macrophages / monocytes were also represented in the top 10 hits as *RGL1* displayed a log2 fold change of 3.74 and p-adj= 2.69e-07.

75 significantly down-regulated hits were recorded with *P3H2* in AT1 cells exhibiting the most significant down-regulation and monocytes / non-classical monocytes being the most represented cell type, occurring in 30% of the top 10 most significant hits.

Investigation of AT2 Cell Type Identity Stability between Conditions

Considering that 338 P53 target genes were significantly differentially expressed in multiple cell types with AT2 cells being most represented amongst the top 10 significant hits, AT2 core marker gene expression were examined between COVID-19 and control conditions to assess AT2 identity stability (fig 15).

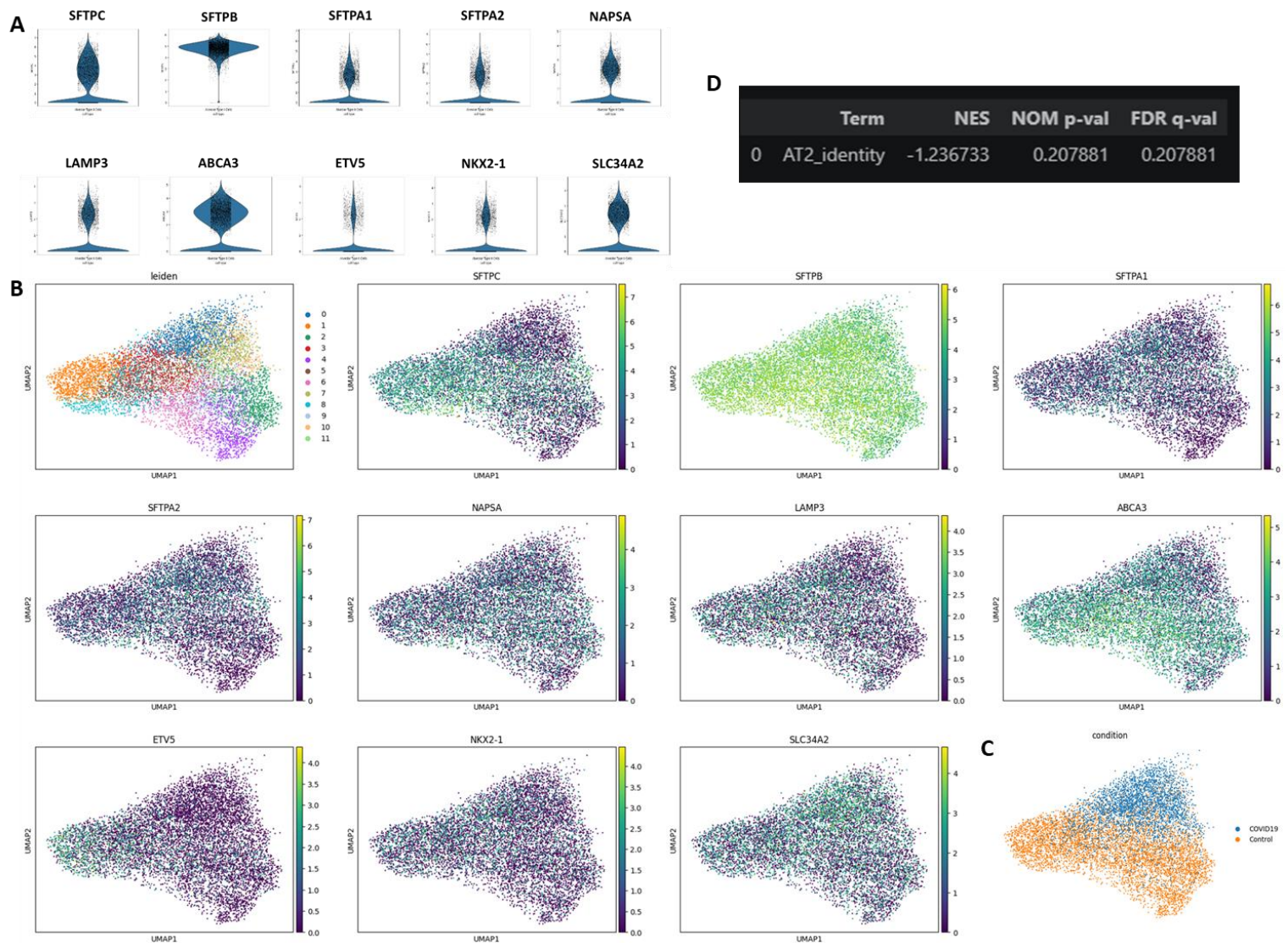


Figure 15. Distribution and expression of AT2 identity markers across the putative “AT2 cell” cluster. (A) Violin plots of core AT2 cell type marker gene expression distribution. (B) AT2 marker spatial gene expression across the AT2 cell cluster. (C) Mapping of AT2 cluster cells by condition. (D) Donor-level GSEA pre-rank score for AT2 identity marker genes.

For the core Alveolar Type II cell marker genes (fig 15A), a pattern of two discrete populations were observed within the Alveolar Type II cluster. All genes except for *SFTPB* there were a greater number of cells exhibiting no expression than the number of cells exhibiting detectable levels of expression.

Interrogation of the AT2 cluster with the Leiden algorithm produced 12 zones of slightly overlapping gene expression from normalised counts (fig 15B). Overlaying AT2 core marker gene expression against the AT2 cell cluster revealed diffuse patterns of varying expression levels across the cluster. Notably, *SFTPC* and *ETV5* exhibited expression concentrated towards the left corner of the roughly triangular cluster while *SFTPA1*, *SFTPA2*, *NAPSA*, *LAMP3* and *NKX2-1* displayed concentrated expression towards the centre of the cluster. *ABCA3*'s expression appeared as a diffuse gradient

concentrated at the bottom centre and left of the cluster, decreasing in intensity towards the top right corner of the cluster. Higher *SLC34A2* expression appeared more concentrated towards the top right of the cluster whereas *SFTPB* expression was close to uniformly highly expressed (> 4) across the cluster.

Overlaying the AT2 cell cluster with the cell condition (fig 15C) revealed a gradient of gene expression overlap between conditions, with control healthy cells concentrated at the bottom and left while COVID-19 cells predominantly localised towards the top right of the triangular cluster. A diffuse overlapping of expression between conditions localised these cells centrally within the cluster.

Analysis of the AT2 identity marker genes via GSEA pre-rank at the donor level (fig 15D) did not produce a significant result but did highlight a general down-regulation of these markers normalised gene expression (NES= -1.236).

Examination of AT1 Marker Expression in AT2 Cells

Surfactant-expressing AT2 cells employ their nascent stem cell functionality by undergoing transcriptional reprogramming to an AT1 phenotype to replenish lost / damaged AT1 populations following acute lung injury or infection (Wu & Song, 2020; Ruaro *et al*, 2021 and Wang *et al*, 2023). To understand whether AT2 cells were undergoing trans-differentiation to AT1 cells, AT2 cells were interrogated for the expression of AT1 marker gene expression.

GSEA pre-rank analysis determined that AT1 marker genes were highly significantly up-regulated in AT2 cells (fig 16A) and that of 22 AT1 marker genes, *MSLN*, *SCEL*, *C19orf33*, *FXVD3*, *AGER*, *KRT19*, *SUSD*, *CAV2*, *SEMA3B*, *CAV1*, *ANKRD1* and *CLIC3* AT1 genes significantly contributed towards AT1 marker expression as leading edge genes (fig 16B).

At the single-cell level, violin plots for the AT2 identity score appear very similar between COVID-19 and control conditions, however more cells are present in the control (fig 17A). A greater proportion of cells are closer to 0 AT2 identity in COVID-19 than in controls and exhibit a greater range in AT2 identity score. A greater proportion of cells express a higher AT1 identity score under COVID-19 conditions relative to the control and the range of AT1 identity score expression is higher under COVID-19 conditions (fig 17B).

The scatterplots illustrate considerable overlap between cells with high and low gene counts within the bulk cluster of cells and with the diffuse break-off cells (fig17C). More cells exhibiting low gene counts (approximately 500) possess higher AT1 and AT2 identity scores than medium gene counts (2000-2500). Additionally, the main cluster cells predominately possess low (0-2%) mitochondrial counts than the break-off cells ($> 4\%$) however some cells with mitochondrial counts greater than 6% also appear in the main cluster (fig 17D). Therefore it is difficult to confirm if cells are outliers, dying, doublets or artefacts by eye.

A

	Term	NES	NOM p-val	FDR q-val	Lead_genes
0	AT1_identity	1.860542	0.000555	0.000555	MSLN;SCEL;C19orf33;AGER;FXYP3;KRT19;SUSD2;CAV2...

B

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	is_leading_edge
MSLN	5.743886	3.732025	0.751674	4.964950	6.871891e-07	0.000043	True
SCEL	67.902079	2.388603	0.488398	4.890692	1.004821e-06	0.000057	True
C19orf33	3.411822	4.331015	0.912581	4.745896	2.075854e-06	0.000096	True
FXYP3	4.882678	3.283851	0.694040	4.731503	2.228636e-06	0.000102	True
AGER	3.568890	4.116121	0.869897	4.731737	2.226068e-06	0.000102	True
KRT19	37.000930	1.524088	0.408338	3.732421	1.896482e-04	0.003097	True
SUSD2	15.707934	2.549142	0.689407	3.697587	2.176587e-04	0.003459	True
CAV2	17.485249	1.649624	0.457145	3.608535	3.079309e-04	0.004480	True
SEMA3B	11.083229	1.887518	0.531594	3.550673	3.842477e-04	0.005258	True
CAV1	17.216131	1.829032	0.646840	2.827643	4.689205e-03	0.032788	True
ANKRD1	1.854477	4.758242	1.781054	2.671588	7.549325e-03	0.045478	True
CLIC3	1.670735	2.333759	0.882272	2.645170	8.164993e-03	0.048150	True

Figure 16. AT1 marker status (A) and leading genes (B) responsible for generating an AT1 transcriptional profile in the AT2 cell cluster population.

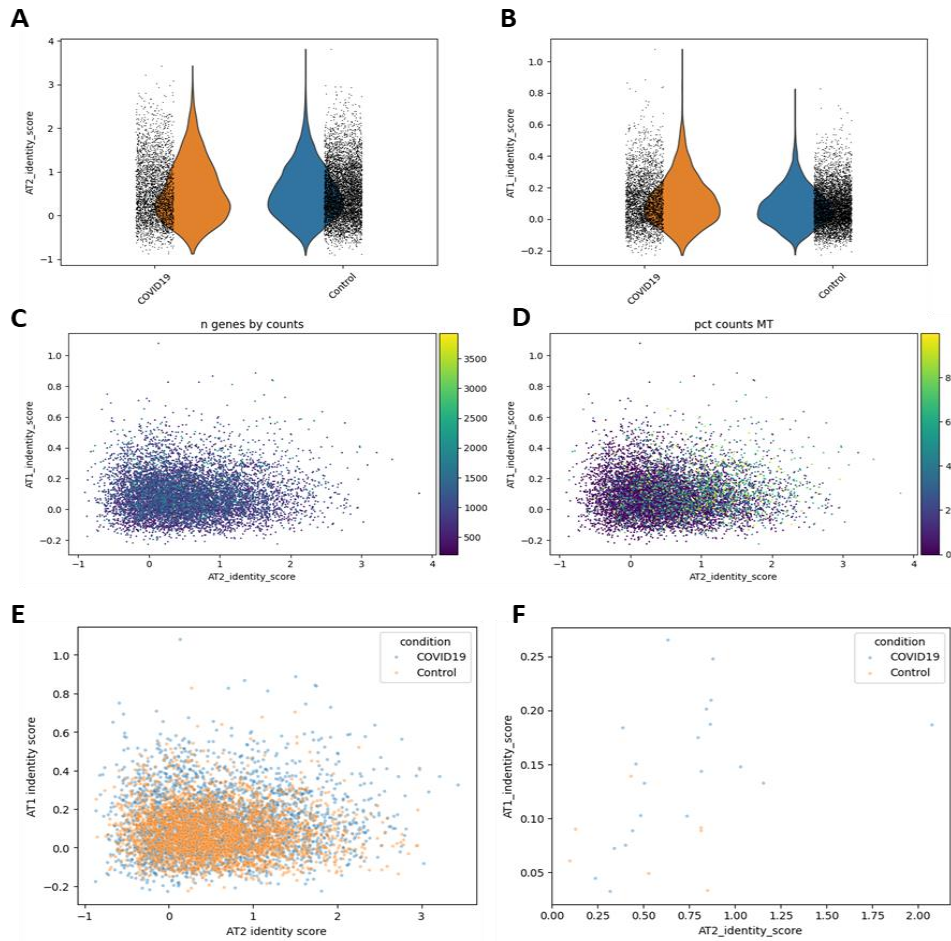


Figure 17. Analysis of AT1-AT2 cell identity distribution at the single cell and donor-levels. (A-B) Violin plots of AT2 and AT1 identity distribution against gene expression levels respectively. (C-D) Scatterplots

of AT1 / AT2 identity distribution by gene counts and percentage mitochondrial gene count respectively. (E) Scatterplot of AT1 / AT2 identity distribution by condition (control and COVID-19 cell counts were equalised). (F) Distribution of AT1 / AT2 identity by donor.

Control AT2 cells were more tightly packed together relative to the COVID-19 cells which were distributed more diffusely in the main cluster bulk (fig 17E). A greater preponderance of COVID-19 cells with higher AT1 identity scores compared to control cells were observed and the highest AT2 identity scoring cells also belonged to COVID-19 condition cells. Therefore overall, control and COVID-19 conditions overlap in AT1 / AT2 identity distribution between cells, however a greater range in AT1 and AT2 identity was observed in single cells under COVID-19 conditions. This observation was replicated at the donor-level (fig 17F) with COVID-19 donors exhibiting greater AT1 and AT2 identity scores relative to control donors.

To assess whether AT1 identity within AT2 cells was a function of donor cell number, the proportion of “AT1 high” cells were defined as those with gene expression signatures in COVID-19 conditions that exceed the 95th percentile of the control cells. “AT_high” frequencies were calculated as a proportion of the total AT2 cell number per donor and plotted (fig 18A).

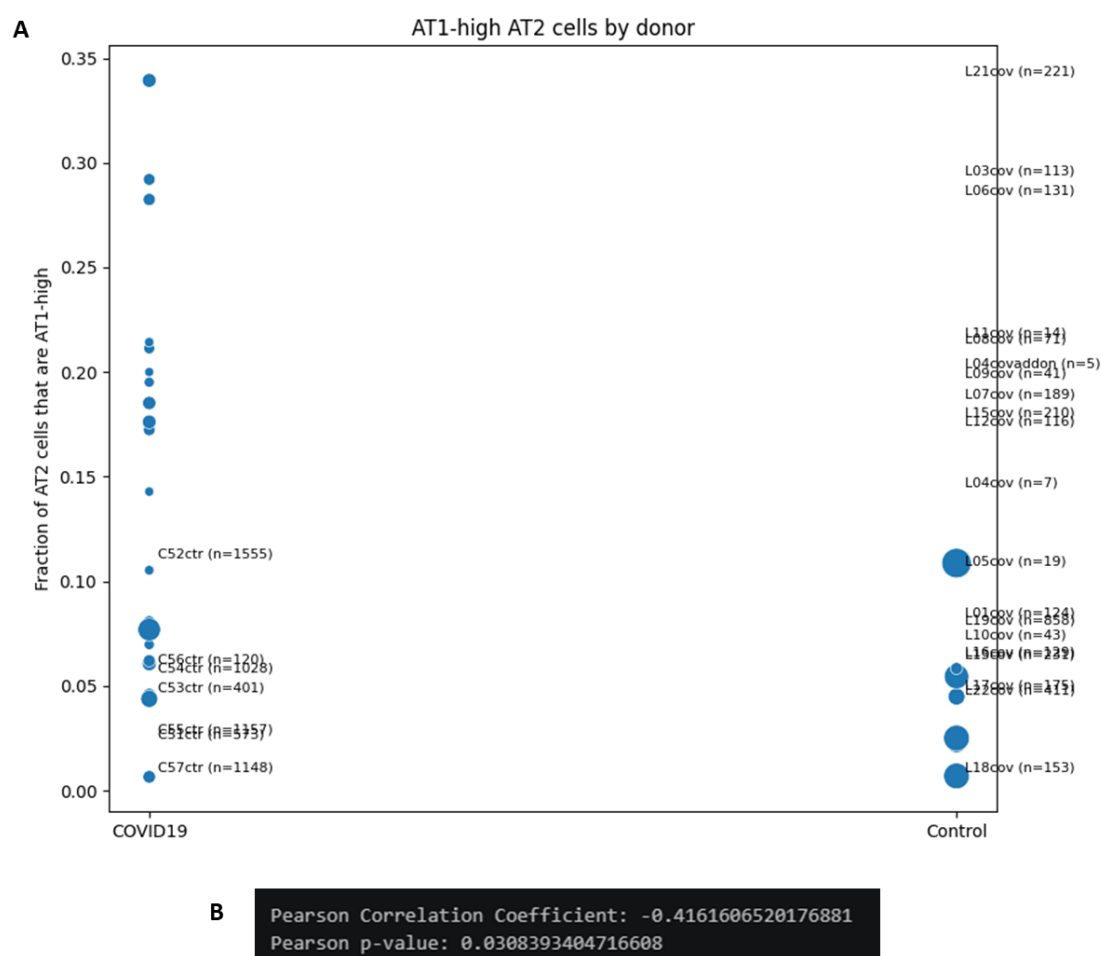


Figure 18. (A) Distribution of AT1 high cells by donor relative to cell number. (B) Pearson Correlation Coefficient and p-value for the effect of donor size and AT1 high proportion.

The plot (fig 18A) clearly illustrates the greater number of high AT1 donors present in the AT2 cells in COVID-19 conditions relative to healthy controls. 11 of the COVID-19 donors exhibited higher “AT1-high” scores than any control donor. The high AT1 identity score weakly negatively correlate with donor cell numbers (fig 18B) ($r = -0.416$, $p = 0.031$) but is significant, indicating that high AT1 identity scores are not independent of donor cell number.

To understand if the AT1 cells are transcriptionally connected to AT2 cells under COVID conditions compared to other lung cell types, partition-based graphical abstraction (PAGA) was performed by interrogating cell connectivities between conditions (fig 19).

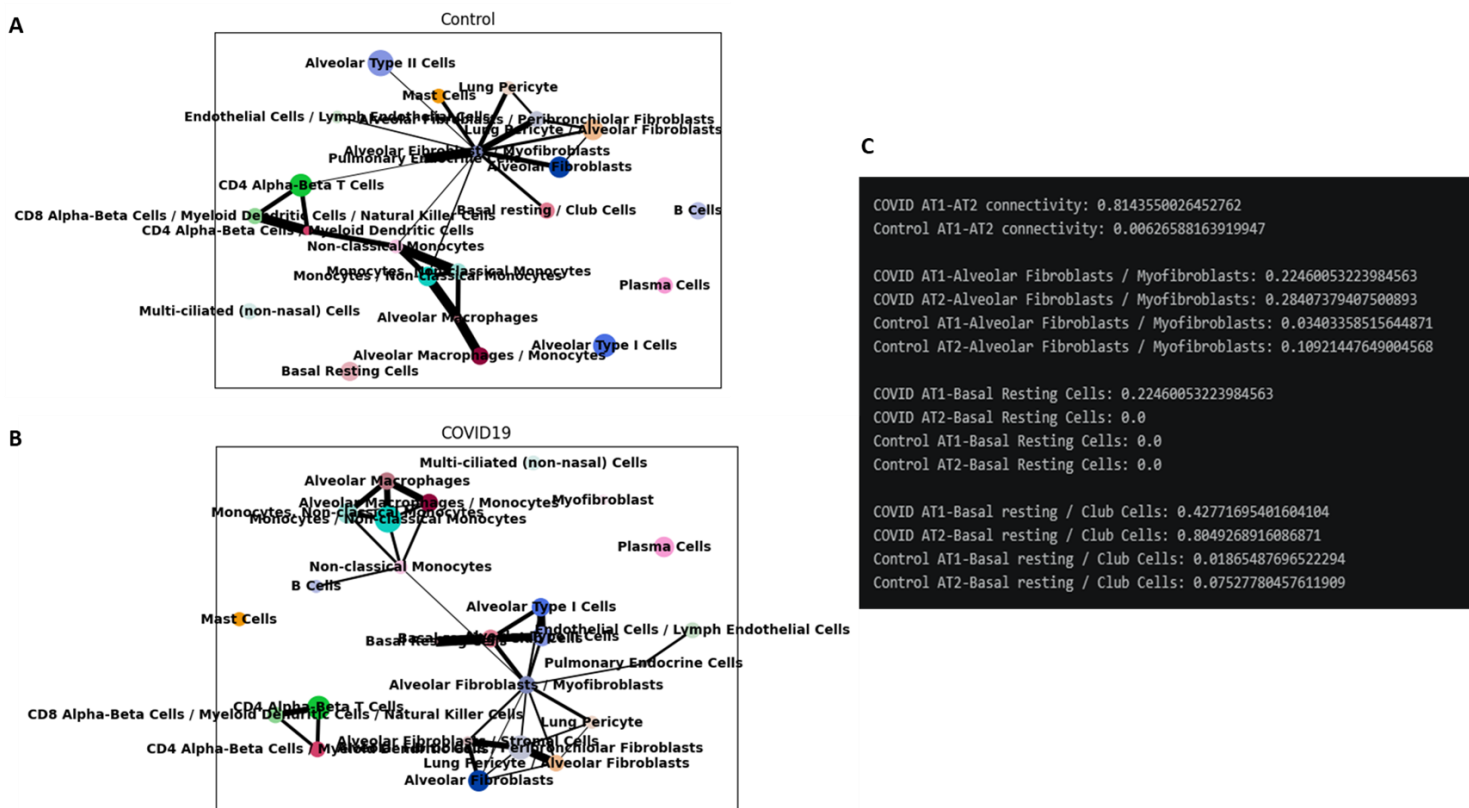


Figure 19. PAGA analysis of connectivities between cell types under control (A) and COVID-19 (B) conditions. Line thickness indicate stronger connections. (C) Quantification of cell type connectivities (larger numbers indicate stronger connections).

PAGA results demonstrate dramatic changes in connectivities across multiple cell types between conditions. Connectivities between AT1 and AT2 cells increased 130-fold under COVID-19 conditions compared to healthy controls from 0.0062 to 0.814. Large-scale increases in connectivity were consistently observed from control to COVID-19 conditions for AT1 - alveolar fibroblast / myofibroblasts (6.58-fold change), AT2 - alveolar fibroblasts / myofibroblasts (2.61 fold change), AT1 - basal resting / club cells (no connectivity to 0.224), AT1- basal resting cells (23.76 -fold increase) and AT2

- basal resting / club cells (10.69). No change in connectivity was observed between AT2 and basal resting cells between healthy and COVID-19 conditions.

Discussion

This analysis of the Melms *et al* dataset revealed dramatic losses in AT1, AT2, lung pericytes and basal resting cells and increases in alveolar fibroblasts / peri-bronchiolar fibroblasts, CD4 positive alpha-beta T cells and monocytes / non-classical monocytes. This is consistent with findings by Delorey *et al*, Rendeiro *et al*, and Szabo *et al* who discovered severe dysregulation of lung epithelial function and immune cell signalling and cytokine mediated inflammatory responses in patients infected with lethal COVID-19 (Delorey *et al*, 2021; Rendeiro *et al*, 2021 & Szabo *et al*, 2021). Additionally, consistent with the results of this work, Delorey *et al* noticed up-regulation of apoptotic pathway genes as well as adaptive immune and inflammatory response pathways (Delorey *et al*, 2021).

This analysis discovered “Ribosome” and “Corona Virus” terms in the top two hits amongst the KEGG gene set following over representation analysis, which may be indicative of viral mRNA reads enriched in AT2 cells and a signify co-option of host ribosomal machinery for Sars-Cov2 viral protein synthesis. This is further supported by reactome results enriched for “Eukaryotic translation elongation”, “Peptide chain elongation”, “SRP-dependent co-translational protein targeting” and “viral mRNA translation.” ORA analysis illustrated “Insulin resistance” and “Circadian rhythm” as highly downregulated which may indicate metabolic reprogramming occurring in AT2 cells and globally in the lung possibly to facilitate viral replication.

Down-regulated and depleted processes discovered by reactome analysis included “AURKA activation by TPX2” as the top hit. Aurora Kinase A (AURKA) is activated by direct binding of TPX2 resulting in steric occlusion of its T288 phosphothreonine group consequent of conformational change induced by TPX2 (Bayliss *et al*, 2003). This prevents deactivation of AURKA by phosphatases, enabling stabilisation and translocation of AURKA to centrosomes to govern centrosome maturation and mitotic spindle assembly (Polverino *et al*, 2024). SARs-CoV2-mediated dysregulation of AURKA-TPX2 controlled spindle formation may suggest an enforced cell-cycle arrest in surviving AT2 cells, preventing proliferation as a defence mechanism (Panda *et al*, 2023).

The Sars-CoV2 virus binding target Angiotensin Converting Enzyme 2 (ACE2) receptors are expressed in pancreatic islet cells as well as AT2 cells, suggesting that dysregulation of islet beta cell function can perturb insulin production and secretion, resulting in insulin resistance (Conte *et al*, 2024). Interestingly, in a study of 143 non-diabetic patients, 30.7% developed long-COVID-19 and 75% of which developed diabetes following acute Sars-CoV2 infection (Man *et al*, 2024). ACE2 receptors are directly impacted by circadian rhythm master regulators *CLOCK* and *BMAL1* in lung tissue (Zhuang *et al*, 2021). While *CLOCK* and *BMAL1* were not among the significantly differentially expressed genes in this dataset, other core circadian rhythm

genes including *PER2*, *CRY1*, *TOX*, *PREX2*, *USP16*, *FILIP1*, *MEGF11*, *STEAP4* and *P3H2* were all significantly downregulated (data not shown).

This analysis uncovered significant upregulation of P53 signalling pathway genes in multiple cell types, particularly AT1 and AT2 cells under COVID-19 conditions. This is commensurate with viral induced p53 signalling to enact p21-mediated G1 phase cell cycle arrest via suppression of CDK4/6 – cyclin D and CDK2 – cyclin E complexes kinase activity (He *et al*, 2005; Zimmerman *et al*, 2006; He *et al*, 2010 & Mai *et al*, 2018). Virally enforced cell cycle arrest facilitates continuous production of viral proteins at the expense of host cell protein synthesis, evidenced by upregulation of “Cellular response to starvation” in our enriched ORA results.

While there was no significant reduction or loss in AT2 “identity”, indicated by down regulation of AT2 marker genes, AT1 core marker genes were upregulated a subset of AT2 cells in COVID-19 donors, suggesting a degree of differentiation occurring in surviving AT2 cells towards AT1 and migration or differentiation of basal resting cells / club cells to AT2 cells, evidenced from PAGA connectivities. This observation may be indicative of maladaptive repair documented to occur following severe lung injury in which discrete populations of p63⁺ stem cells migrate to sites of AT1 ablation and express basal cell markers such as *KRT5* and club cell markers including *SCGB3A2* and *CYP2F2* in failing to reconstitute alveolar morphology. Chronic pulmonary dysfunction ensues as a result (Kumar *et al*, 2011; Vaughan *et al*, 2014 & de Mello Costa *et al*, 2020).

In summary, SARs-CoV2 infection dysregulates multiple signalling pathways including protein synthesis, host cell innate and adaptive immune responses, cellular proliferation, apoptosis, glucose metabolism, circadian rhythm and stem cell mediated alveolar replenishment. These pathways might serve as possible targets for attenuating viral attack, specifically down-regulating p53-mediated cell cycle arrest via p21 to maintain cell proliferation, stabilising *PER2* and *CRY1* expression to ensure BMAL1-CLOCK regulation of circadian rhythm cycles and controlled / conditional deactivation of *AURKA* to prevent G1/S or G2/M phase cell cycle arrest.

Limitations

The onset of lethality in COVID-19 patients can produce vastly different physiological outcomes depending on the speed of symptom presentation. Severe hypoxia and rapid clinical deterioration without any visible presentation of acute lung injury are commonly associated with early onset of lethality in respiratory disease compared to the detection of pulmonary immune infiltrates and identification of histological markers of acute lung infection which are hallmarks of late onset of lethality (De Michele *et al*, 2020; Herrmann *et al*, 2020; Nienhold *et al*, 2020 & Yoon *et al*, 2020). The authors, Melms *et al* do not specify if COVID-19 donors were stratified by early or late onset of death following hospitalisation, therefore vast differences in lung cell composition between donors might be skewing results. Moreover, the high viral load present in

early onset of death donors can significantly enrich viral RNA reads at sequencing depths which wouldn't detect them in late onset donors.

Additionally, the number of donors between conditions is not equal. While not problematic if given sufficient cell population sizes, our analysis has reflected instances where specific cell types e.g. AT2 cells may be orders of magnitude fewer in COVID-19 conditions relative to controls. These minute sample sizes can inflate statistical power of results owing to high noise within the sample that would generally dissipate when sample size increases. Elevating the minimum cell threshold can compensate for this effect but can incur the trade-off of donor exclusion and hence biological replicate reduction.

While the authors instantiate a rigorous post-mortem protocol by harvesting organs and tissue within a 10 hour window of death to preserve RNA quality, discrepancies inevitably arise between live and post-mortem samples which can affect cell number and RNA quality between conditions.

Precise and efficient mapping of cell type identities to cell clusters is imperative for dataset interrogation and results interpretation. While multiple independent database and literature resources were consulted to corroborate cell specific markers and validate predictions via Cell Typist, Cell Typist cannot accurately distinguish between healthy and diseased tissue types and thus discrepancies in cell type label assignment are difficult to deconvolute.

Considering circadian rhythms were investigated in this work and that ORA analysis identified their significant down-regulation in COVID-19 conditions, awareness of their rhythmic expression patterns should be noted. Regulatory components expression of the circadian rhythm machinery including *PER2*, *CRY1*, *CLOCK* and *BMAL1* may fluctuate depending on the time of day (Vitaterna *et al*, 2001), implying that donor sample harvest time should be accurately time stamped to ensure comparability between samples. Unfortunately, this information was not stated by Melms *et al*.

Melms *et al* included lung biopsies and lung resections from healthy donors and post-mortem tissue from lethal COVID-19 donors. While outside the scope of their work, with considerations of posterity, having access to lung biopsies from COVID-19 survivors would increase the resolution of cell type specific transcriptional changes, population dynamics and adaptation to viral infection.

Future Work

Notable observations that eluded in-depth exploration due to time and scope constraints included investigation into basal cell / club cell reprogramming to AT1 and AT2 as well as fibrotic remodelling of the alveolar niche indicated by the massive increase in alveolar / peri-bronchiolar fibroblasts under COVID-19 conditions. Additionally, the infiltration, transcriptional profile and distribution of monocytes / non-classical monocytes would provide key insights into the innate immune response in the lung. The incorporation of genome wide association (GWAS) study datasets would facilitate the examination of SNP profiles which, though techniques such as express

quantitative trait loci (eQTL), might reveal regulatory genomic regions endowing susceptibility or resistance to SARs-CoV 2 infection at the population level. In terms of wet-lab studies, Bi-directional CRISPRa / CRISPRi Perturb-Seq screens could be performed with a SARs-CoV 2 challenge to identify genes conferring susceptibility or resistance to infection and thus potential druggable targets.

Data

The original datasets used in this project are accessible from:
<https://www.nature.com/articles/s41586-021-03569-1#data-availability>

Code

The Python code used to perform this analysis can be found at:
<https://github.com/AshleyRamsawhook>

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